# Vishnu Nithin Reddy

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## Education

- **Indian Institute of Information Technology** — Lucknow, UP
  *Bachelor of Technology in Computer Science and Artificial Intelligence; GPA: 8.2* — *Dec. 2020 – Jun. 2024*

## Experience

- **BuildShip** *(A powerful low-code visual back-end builder.)*
  *Engineer* — *Oct 2023 - Present*
  - Building user-facing features using React, Jotai, TypeScript, Material UI & Firestore. Built a fully-responsive public-facing BuildShip Templates & Nodes explorer **NextJS** app from start-to-finish, leveraging ISR and using Tailwind for styling. App scores a perfect 100 on 3 of the 4 Lighthouse categories.
  - Built the Global Search module using **Meilisearch** (deployed to a VM instance on GCP) from scratch, indexing a total of about 16000 workflow documents with periodic syncing handled via CRON (changed documents, every 2 minutes) serving to hundreds of concurrent users across thousands of total tenants.
  - Migrated the core project to a **Turborepo**-managed monorepo with multiple apps & shared packages (shared UI components, Tailwind config & icon-component-collections as self-contained packages consumed by Vite and NextJS apps), with clean HMR across package boundaries.
  - Built a Storage settings interface (a file explorer) on top of **Google Cloud Storage**'s flat namespace to allow users to interact with and manage their project storage. Created APIs using **Multer** to allow for file uploads on Cloud Run instances (with form-parsing, per-execution-isolated storage and post-execution cleanup).
  - Improved the app's lazy-loading, bundling, and chunking strategies, and reduced the built entrypoint bundle size by ∼48% (FCP down by 1.1s, blocking time down by 1.2s, among other improvements) & managed preview deployments on Firebase Hosting via GitHub Actions workflows.

- **Zenskar** *(An early-stage B2B SaaS startup, solving billing.)*
  *Engineer (Intern)* — *April 2023 - Sep 2023*
  - Built user-facing pages using React, RecoilJS, TypeScript, Tailwind & Material UI, including V0 of the UI for the Simple Contract-Template Editor (one of Zenskar's core modules). Set up and extended a standard design system built upon **Storybook** for use across all of Zenskar for improved DX.
  - Integrated **Vitest** into Zenskar's front-end repository, along with a GitHub Actions Workflow with dependency-caching for quick under-a-minute CI test runs.
  - Reconfigured the Cypress test-suite to reduce memory usage by 13% and CPU usage by 75%. Employed dependency-caching in the Actions workflow to lower CI test run times from ≈ 9 minutes/run to ≈ 3.5 minutes/run.

- **Google Summer of Code 2022** *(An open-source program hosted annually by Google.)*
  *Open-source Contributor – worked with Oppia Foundation* — *June 2022 - Sep 2022*
  - Created UI mocks in collaboration with designers from the core Oppia team in Figma ***(see #1 and #2)***
  - Introduced multiple Angular components with custom SVG animations built from scratch, a small API to fetch lesson stats written in Python, and corresponding Jasmine, Protractor & WebDriverIO tests to catch regressions.

- **Oppia Foundation** *(An open-source project aimed at making engaging education accessible.)*
  *Open-source Collaborator & Org Member* — *Late 2021 - Mar 2023*
  - Worked extensively with Angular, RxJS, TypeScript and introduced several new components, services and bug-fixes, alongside Jasmine and Protractor tests. Eventually co-led the team managing one of the core modules of Oppia.

## Projects

- **AudioPass**: Open-source multi-threaded Linux desktop app that lets you play local audio through a virtual mic; UI and audio processing handled by separate dedicated threads communicating via MPSC and PipeWire channels.
- **Open Preview Peek**: Open-source browser extension to preview a webpage's Open Graph and Twitter link metadata; reads meta tags via content script and sends data back to popup through background script (each with its own Vite config). Ideal for use with local web app development.

## Skills

- **Languages**: TypeScript, Rust.
- **Technologies**: React, Jotai, Firebase/Firestore, NextJS, TailwindCSS, KoaJS, Material UI, egui, GitHub Actions, Storybook. Familiar with GCP and worked in the past with Angular, Python, Jasmine/Cypress/Vitest.